A Consistent Theory of Coded SP-AFDM Reception

Fractional-Doppler representation, statistical inference, and computation-limited receiver design

Technical manuscript and reproducible verification package

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1 Scope, status, and provenance

This chapter gives a self-contained theoretical treatment of joint channel estimation, detection, and decoding with superimposed pilots (SP) in an affine frequency division multiplexing (AFDM) system. It preserves an integer delay-Doppler receiver dictionary while allowing fractional Doppler in the propagation model. It does not require continuous Doppler estimation.

The mathematical statements below are proved under explicitly stated assumptions. Their implementations are checked by independent finite-dimensional calculations, exact finite-alphabet enumeration, deterministic quadrature, and Monte Carlo experiments. The accompanying script passes **50 numerical assertions**, organized into **20 labeled verification groups**. Numerical agreement is a consistency check, not a replacement for a proof.

Evidence boundary. The new experiments are independently implemented reference models, not a replay of the project's production I1/LDPC receiver. The available project materials include historical audits and the September 12 reassessment, but not a complete, executable, version-matched production environment with its raw confirmation arrays. No new production FER, production timing, state-of-the-art superiority, or publication novelty is asserted. The central model and its conditional theory can be used in the thesis; receiver-level superiority remains an empirical question.

The project-derived basis is the September 12 reassessment [P1], supported by its report template [P2]. The September 11 summary [P3] and July audits [P4] are treated as earlier evidence, not as instructions or unquestionable theoretical conclusions. In particular, the September reassessment explicitly distinguishes exact identities from unconfirmed FER gains. Some earlier stronger interpretations are corrected in Section 12.

The literature establishes the AFDM operator model [R1], iterative SP-AFDM estimation/detection [R2], SP message-passing methods [R3], coded SP feedback [R6], and direct AFDM-SP off-grid sparse Bayesian methods [R7]. Accordingly, neither joint reception, soft variance, whitening, nor combining these ingredients is claimed as a new invention here.

2 Problem formulation and signal model

2.1 Objective and notation

Let b denote an information message, $d(b) \in \mathbb{C}^N$ its coded and modulated data vector, and $p \in \mathbb{C}^N$ a deterministic known pilot. Both vectors already include their transmit-power scaling. Define

$$x = p + d(b), \qquad y = H(\theta)x + w, \qquad w \sim \mathcal{CN}(0, \sigma^2 I_N). \tag{1}$$

The convention is $\mathbb{E}|w_n|^2 = \sigma^2$; each real component has variance $\sigma^2/2$. For the project's scaled QPSK alphabet $\{\pm 0.5 \pm 0.5j\}$, the data energy is 0.5 per symbol, not one. A variance computed from this alphabet must not be multiplied by the data-power fraction again [P1, Assumptions and Main Derivation 1].

For a prescribed receiver algorithm $\mathcal{A}$, the operational objective is

$$\min_{\mathcal{A}} R_H(\mathcal{A}), \qquad R_H(\mathcal{A}) = \Pr_H\{\hat{b}_{\mathcal{A}}(y) \neq b\}, \qquad \mathbb{E}[C(\mathcal{A})] \leq B. \tag{2}$$

Here C includes initialization, channel estimation, detection, decoder calls, covariance acquisition, and message handling. Averaging R_H over a declared channel law gives the channel-ensemble risk. Channel NMSE, residual energy, and soft-output log loss are diagnostics, not interchangeable definitions of (2).

Use P for physical path count, S for active integer coefficient count, and Q for total receiver dictionary size. These quantities must not be identified with one another. All vector indices are zero-based and circular subscripts are modulo N.

2.2 Physical assumptions

The principal propagation model assumes integer sample delays $0 \leq l < N$, path gains and Doppler frequencies constant over one useful block, an adequate chirp-periodic prefix, ideal synchronization, and additive proper white Gaussian noise after sampling. Fractional Doppler is retained exactly. Fractional delay, arbitrary pulse-shaping mismatch, oscillator impairments, inter-block channel changes, and nonlinear power amplification are outside this principal model.

These assumptions define an ideal sampled linear time-varying channel, not every physical AFDM implementation. A change of pulse shape or receiver filter requires a corresponding change of the sampled operator and possibly the noise covariance. SP removes pilot-only resource separation; it does not remove the physical prefix requirement.

2.3 Exact DAFT and prefix operators

Define the unitary DFT and chirp matrices by

$$F_{mn} = N^{-1/2} e^{-j2\pi mn/N}, \qquad \Lambda_c = \mathrm{diag}(e^{-j2\pi cn^2}), \qquad A = \Lambda_{c_2} F \Lambda_{c_1}. \tag{3}$$

The time-domain block is $s = A^H x$. Let $[\Pi_l s]_n = s_{n-l}$ and

$$\Delta_\nu = \mathrm{diag}(e^{-j2\pi\nu n/N}), \qquad [C_l]_{nn} = \begin{cases} e^{-j2\pi c_1[N^2-2N(l-n)]}, & n < l, \\ 1, & n \geq l. \end{cases} \tag{4}$$

This fixes the negative Doppler-exponential convention; a positive-exponential convention reverses the Doppler signs consistently. The path operator and channel are

$$U(l,\nu) = AC_l\Delta_\nu\Pi_l A^H, \qquad H = \sum_{r=1}^{P} h_r U(l_r,\nu_r). \tag{5}$$

The prefix phase is retained even when special parameter choices make it equal to one. All factors of each path operator are unitary, so $U(l,\nu)U(l,\nu)^H = I$. Also $Aw \sim \mathcal{CN}(0,\sigma^2 I)$. These normalization and prefix details are essential to a faithful implementation of the established AFDM construction [R1].

Proposition 1 (explicit input-output relation). Define

$$D_N(t) = \frac{1}{N}\sum_{n=0}^{N-1} e^{j2\pi nt/N}.$$

Then

$$[U(l,\nu)]_{mk} = e^{j2\pi[c_1 l^2 - kl/N + c_2(k^2-m^2)]} D_N(k-m-\nu-2Nc_1 l). \tag{6}$$

Proof. Insert (3)-(4) into (5). The prefix phase cancels the additional quadratic phase caused by replacing $n-l$ with $(n-l) \bmod N$. The remaining quadratic difference is $(n-l)^2 - n^2 = l^2 - 2nl$. The terms independent of n give the exponential in (6), and the sum of the remaining exponentials gives D_N. This derivation also fixes all signs and the $1/N$ factor.

If $\alpha = 2Nc_1$ and $\nu = q$ are integers, $D_N(k-m-q-\alpha l)$ is nonzero only when

$$k = m + q + \alpha l \pmod N. \tag{7}$$

Thus an integer atom is a phase-weighted permutation. In the notation $[U_i x]_n = u_i[n] x_{n-s_i}$, its effective shift is $s_i = -(q_i + \alpha l_i) \bmod N$ and $|u_i[n]| = 1$. Distinct physical coordinates do not automatically imply distinct effective shifts; injectivity over the actual dictionary must be checked.

Verification. T01 compares direct prefix/time-domain construction, the explicit formula, unitary identities, and FFT-based application, including a nontrivial prefix parameter choice. Maximum discrepancy: 1.21×10^{-14}.

3 Fractional Doppler and integer-domain representation

3.1 Exact finite expansion

Write $\nu = k_0 + \delta$, $\delta \in [-1/2, 1/2)$. Integer Doppler means finite-block grid alignment, not low mobility.

Proposition 2 (fractional Doppler does not require continuous estimation). Let

$$c_q(\nu) = \frac{1}{N} \sum_{n=0}^{N-1} e^{-j2\pi(\nu-q)n/N} = D_N(q-\nu).$$

Then

$$\Delta_\nu = \sum_{q=0}^{N-1} c_q(\nu)\Delta_q, \qquad U(l,\nu) = \sum_{q=0}^{N-1} c_q(\nu)U(l,q), \qquad \sum_q |c_q(\nu)|^2 = 1. \tag{8}$$

Proof. Equation (8) is the inverse finite Fourier expansion of the sampled sequence $e^{-j2\pi\nu n/N}$. Multiplication by the fixed matrices A, C_l, Π_l, A^H preserves the equality. Parseval's identity gives the coefficient-energy relation.

For integer ν, one coefficient is nonzero. For noninteger ν, the complete expansion is generally dense. A small physical path count therefore does not imply an equally small integer coefficient count. Conversely, a complete integer expansion represents the sampled channel exactly; continuous estimation is one compact parameterization, not a mathematical necessity.

3.2 Orthogonality, merging, and truncation

Proposition 3 (operator-space orthogonality). For distinct pairs (l,q) with $l, q \in \{0, \dots, N-1\}$,

$$\mathrm{tr}[U(l,q)^H U(l',q')] = N\, \mathbf{1}_{l=l'} \mathbf{1}_{q=q'}. \tag{9}$$

Proof. Unitary conjugation by A preserves the trace inner product. If $l \neq l'$, the product of the time-domain operators is a weighted nonzero circular shift and has zero diagonal, hence zero trace. If $l = l'$, the prefix factors cancel and the trace reduces to the sum of an integer Doppler exponential, equal to N when $q = q'$ and zero otherwise.

For multiple physical paths, first combine contributions that share the same integer pair:

$$g_{lq} = \sum_{r:l_r=l} h_r c_q(\nu_r), \qquad H = \sum_{l,q} g_{lq} U(l,q). \tag{10}$$

For an index set $\mathcal{S}$, define $H_{\mathcal{S}}^F = \sum_{(l,q)\in\mathcal{S}} g_{lq} U(l,q)$. By (9),

$$\|H - H_{\mathcal{S}}^F\|_F^2 = N \sum_{(l,q)\notin\mathcal{S}} |g_{lq}|^2. \tag{11}$$

This is the Frobenius-orthogonal projection onto the retained operator subspace. Selecting the largest merged coefficient energies minimizes (11) at fixed cardinality, but requires true coefficients and is

therefore an oracle diagnostic. Different paths with the same delay must be merged before their coefficient energies are squared. Summing separate path-tail energies is generally incorrect because their coherent cross terms do not vanish in an individual channel.

For one path, retaining a set $\mathcal{K}$ leaves the exact fraction

$$\epsilon_{\mathcal{K}}(\nu) = 1 - \sum_{q \in \mathcal{K}} |c_q(\nu)|^2. \tag{12}$$

At $N = 64$, half-bin offset, and retention of the largest-magnitude coefficients:

Retained coefficients	Retained path energy	Omitted path energy
1	40.5366%	59.4634%
2	81.0732%	18.9268%
5	91.7251%	8.2749%
9	95.5336%	4.4664%
17	97.7505%	2.2495%
33	99.0522%	0.9478%
64	100%	Numerical zero

These are operator-energy calculations, not channel-estimation or FER results. Truncation is not followed by renormalization: rescaling retained coefficients cannot restore omitted phases or couplings.

Verification. T02 checks (8)-(12). T20 checks coherent merging in a multipath channel: exact omitted operator energy 1.5557469 equals the merged-coefficient result, whereas an incorrect pathwise energy sum gives 1.0044525.

A full operator dictionary has N^2 coefficients. Exact representability does not imply those coefficients can be identified from one N-sample observation. Sparsity, physical structure, coding, and repeated observations remain essential restrictions.

4 What superimposition makes difficult

4.1 No universal linear separation

Proposition 4. If the allowed data vectors span $\mathbb{C}^N$, any fixed linear map L satisfying $LHd = 0$ for every allowed d also satisfies $LHp = 0$.

Proof. Linearity implies $LH = 0$ on the span of all allowed data vectors, which is the whole input space. Apply this identity to p.

Exact pilot-data separation therefore needs extra structure. Statistical averaging, transmitter-side data restrictions, spatial separation, or decoder information are different ways to supply it. This proposition concerns universal linear cancellation; it is not an impossibility theorem for nonlinear joint inference.

Even noiseless joint identification needs care. For one scalar observation and an unrestricted unknown gain, each nonzero candidate transmit symbol can be fitted by a different gain. A known superimposed pilot alone does not guarantee unique recovery of both gain and data. For a known data vector and fixed channel support, coefficient identifiability reduces to the rank of the corresponding training

matrix. Full-rank pilot responses identify the channel from its conditional mean, but rank deficiency of that mean alone does not prove nonidentifiability of the complete channel-dependent distribution.

4.2 A minimal flat-fading calculation

For independent zero-mean unit-variance symbols z_n,

$$y_n = h(\sqrt{P_p} + \sqrt{P_d} z_n) + w_n, \qquad \hat{h} = \frac{1}{N\sqrt{P_p}} \sum_n y_n.$$

Then

$$\mathbb{E}[|\hat{h} - h|^2 \mid h] = \frac{P_d |h|^2 + \sigma^2}{N P_p}. \tag{13}$$

Proof. Subtract h and expand the two independent zero-mean sums. The cross terms vanish and each sum contributes N times its single-sample variance.

At fixed block length and pilot-data ratio, the data term remains when thermal noise vanishes. This is a floor of this correlation estimator, not a universal SP error floor. Finite-alphabet or code-aware estimators exploit information discarded by this estimator. Classical iterative SP methods [R2-R3] are motivated by that distinction.

Verification. T17 uses $N = 32$, $P_p = 0.4$, $P_d = 0.6$, $|h| = 1$, $\sigma^2 = 0.1$, and 100,000 independent blocks. Theory predicts 0.0546875; simulation gives 0.0545319, with Monte Carlo standard error 0.0001697.

5 Exact nuisance moments and the limits of Gaussian whitening

5.1 Conditional versus channel-averaged covariance

For initial data statistics $\mathbb{E}[d] = 0$ and $\mathrm{Cov}(d) = V_0$,

$$\eta = Hd + w, \qquad C_\eta(H) = H V_0 H^H + \sigma^2 I. \tag{14}$$

This is an exact second-moment identity. It does not assume Gaussian data. The commonly used special case is $V_0 = P_d I$, but code constraints can introduce off-diagonal covariance; the actual codebook or declared approximation determines V_0.

For $H = h_0 I + h_1 \Pi_1$, HH^H includes shifted cross terms $h_0 h_1^* \Pi_1^H + h_1 h_0^* \Pi_1$. Static integer-delay multipath can therefore produce colored data interference. For one ideal path $H = hU$ with unitary U, $HH^H = |h|^2 I$, so fractional Doppler can produce white covariance.

If path gains are mutually uncorrelated, zero mean, and independent of isotropic data, averaging over those gains removes cross-path terms:

$$\mathbb{E}_H C_\eta(H) = \left(P_d \sum_r \mathbb{E}|h_r|^2 + \sigma^2 \right) I. \tag{15}$$

Equations (14) and (15) condition on different information. Their coexistence explains why an ensemble-white model, such as the statistical formulation in [R2], need not describe each realized frame.

Verification. T03 verifies (14) by enumerating all 256 QPSK vectors for $N = 4$. T03b confirms white covariance for a fractional-Doppler single path.

5.2 Non-Gaussianity does not disappear merely because N is large

Proposition 5 (fourth cumulant of QPSK interference). For independent proper QPSK symbols with energy P_d and $z = \sum_n a_n d_n$,

$$\mathbb{E}|z|^4 - 2(\mathbb{E}|z|^2)^2 = -P_d^2 \sum_n |a_n|^4. \tag{16}$$

Proof. Independence and zero mean leave only paired terms in the fourth-moment expansion. Properness eliminates products involving $\mathbb{E}[d_n^2]$. A proper complex Gaussian would contribute $2P_d^2$ per individual fourth moment, whereas constant-modulus QPSK contributes P_d^2. The difference is the negative term in (16).

Define effective mixing

$$M_{\text{eff}} = \frac{(\sum_n |a_n|^2)^2}{\sum_n |a_n|^4}.$$

The normalized fourth cumulant is $-1/M_{\text{eff}}$, not automatically $-1/N$. For one full fractional-Doppler AFDM path with integer $2Nc_1$, the row magnitudes are a permutation of $|c_q(\nu)|$.

Proposition 6 (exact fractional-path fourth moment).

$$\sum_q |c_q(k_0 + \delta)|^4 = 1 - \frac{2}{3}\left(1 - \frac{1}{N^2}\right)\sin^2(\pi\delta). \tag{17}$$

Proof. Set $z_n = e^{-j2\pi\nu n/N}$. Its normalized circular autocorrelation at lag k is

$$r_k = e^{-j2\pi\nu k/N}\left(\frac{N-k}{N} + \frac{k}{N}e^{j2\pi\nu}\right).$$

Hence $|r_k|^2 = 1 - 4\sin^2(\pi\nu)k(N-k)/N^2$. Fourier Parseval gives $\sum_q |c_q|^4 = N^{-1}\sum_k |r_k|^2$. Finally $\sum_{k=0}^{N-1} k(N-k) = N(N^2-1)/6$, yielding (17).

At half-bin offset and large N, $M_{\text{eff}} \to 3$ and the normalized fourth cumulant tends to $-1/3$, not zero. At integer Doppler it equals -1. Thus a large block does not justify a Gaussian approximation in this single-path example. Additional diffuse multipath or dominant thermal noise may improve an approximation, but that requires a separate argument.

Adding independent proper Gaussian noise leaves the fourth cumulant numerator unchanged while increasing the variance denominator. An invertible unitary transform also cannot turn an exactly discrete joint distribution into a Gaussian joint distribution.

Verification. T04 checks finite-alphabet moments; T04b checks (17) for twelve (N, δ) settings. At $N = 16, 64, 1024$ and half-bin offset, $M_{\text{eff}} = 2.97674, 2.99854, 2.99999$.

5.3 Truncated tails are not independent white noise

Write $H = H_R + H_T$. Then the received tail is

$$H_T p + H_T d,$$

with a coherent pilot component. For isotropic data,

$$\mathrm{Cov}(H_R d, H_T d) = P_d H_R H_T^H. \tag{18}$$

Frobenius orthogonality $\mathrm{tr}(H_R^H H_T) = 0$ does not imply the matrix product in (18) vanishes. Expanding HH^H requires both cross-covariance terms.

T08 uses one half-bin path, $N = 16$, and its two strongest integer coefficients. The omitted cross-term matrix has Frobenius norm **42.22%** of the complete data covariance norm, although the retained and omitted operators are Frobenius-orthogonal to numerical precision. Replacing that tail with independent noise is therefore a substantive approximation.

5.4 Marginal and complete-data formulations must not be mixed silently

If the data are deliberately approximated as Gaussian and integrated out, the marginal model is

$$y \mid g \dot{\sim} \mathcal{CN}(H(g)p, C(g)), \qquad C(g) = H(g)V_0 H(g)^H + \sigma^2 I.$$

Its negative log likelihood contains

$$(y - H(g)p)^H C(g)^{-1}(y - H(g)p) + \log\det C(g). \tag{19}$$

Because C depends on g, minimizing only a plug-in weighted residual is not exact marginal maximum likelihood. In contrast, a complete-data method conditions on d and retains thermal covariance $\sigma^2 I$ in (1), then averages the complete-data loss over a working distribution. Section 7 derives that alternative.

Applying a marginal data-interference whitening matrix and then independently adding the complete-data uncertainty penalty is not automatically the same Bayesian calculation. It can be defined as a heuristic weighted objective, but requires its own derivation and interpretation.

6 Support evidence after nuisance elimination

6.1 The correct fixed-covariance, fixed-support test

Consider the deliberately simplified experiment

$$y = \Phi_{\mathcal{S}} a + \phi_j h + \eta, \qquad \eta \sim \mathcal{CN}(0, C), \quad C \succ 0. \tag{20}$$

The current support $\mathcal{S}$ and C are held fixed and the current coefficients a are unknown nuisance parameters. Let $C = LL^H$. Choose a matrix Z with orthonormal columns spanning the orthogonal complement of range$(L^{-1}\Phi_{\mathcal{S}})$. Define

$$z = Z^H L^{-1} y, \qquad v_j = Z^H L^{-1} \phi_j. \tag{21}$$

The projected experiment is $z = v_j h + \xi$ with $\xi \sim \mathcal{CN}(0, I)$. If $v_j = 0$, the candidate adds no identifiable direction beyond the existing support and should not be assigned a finite normalized evidence score.

Proposition 7 (projected evidence). For $v_j \neq 0$, maximizing over the unknown complex h gives the log generalized likelihood ratio

$$T_j = \frac{|v_j^H z|^2}{v_j^H v_j}. \tag{22}$$

If instead $h \sim \mathcal{CN}(0, \tau_j^2)$, the Bayes factor for this projected experiment is

$$\log B_j = \frac{b_j}{1 + b_j} T_j - \log(1 + b_j), \qquad b_j = \tau_j^2 v_j^H v_j. \tag{23}$$

The posterior log odds are $\log[\pi_j/(1 - \pi_j)] + \log B_j$ for prior existence probability π_j.

Proof. The least-squares maximizer is $\widehat{h} = v_j^H z/(v_j^H v_j)$, and the decrease of squared residual is (22). For the random-gain model, $z \mid H_1$ has covariance $I + \tau_j^2 v_j v_j^H$. The matrix determinant lemma and rank-one inverse formula give (23).

The Bayes statement is exact for the projected experiment. A full Bayesian treatment of the current coefficients, rather than eliminating their subspace, requires specifying and integrating their prior as well.

6.2 Why the refitted test is not generally OMP

With white noise, write $M_{\mathcal{S}} = I - \Phi_{\mathcal{S}} \Phi_{\mathcal{S}}^{\dagger}$. The exact refitted residual decrease is

$$\frac{|\phi_j^H M_{\mathcal{S}} y|^2}{\|M_{\mathcal{S}} \phi_j\|^2}. \tag{24}$$

Ordinary OMP uses correlation with the residual, usually after normalizing the original dictionary columns. Orthogonal least squares (OLS) normalizes the candidate after projection and selects the largest actual refitted decrease. These are distinct greedy algorithms [R5].

A three-dimensional counterexample makes the distinction explicit. Let the existing support be e_1, residual $r = e_2$, and unit-norm candidates

$$a_1 = \sqrt{0.99}\, e_1 + 0.1 e_2, \qquad a_2 = 0.8 e_2 + 0.6 e_3.$$

OMP correlation energies are 0.01 and 0.64, so it selects a_2. The refitted residual decreases are 1 and 0.64, so OLS selects a_1. No statistical approximation is involved in this counterexample.

At an empty support and equal dictionary-column norms, the white-noise GLRT ranking reduces to the usual OMP ranking. At later steps this equivalence needs equal projected norms. For the complete Bayesian ranking in (23), flat existence priors and equal b_j are also needed; white covariance or flat priors alone are insufficient.

6.3 Projection, selection, and plug-in qualifications

For a deterministic projection M, the covariance of a projected disturbance is exactly MCM^H, not C. However, if M is selected from the same observed data, conditioning on that selected support changes the disturbance law. Treating the chosen support as a fixed design is then another approximation.

Similarly, (20) holds C fixed across the two hypotheses. In physical SP reception, adding a channel component also changes the distribution of the unknown data contribution. A score using a frozen $\widehat{C} = \widehat{H}V_0\widehat{H}^H + \sigma^2 I$ is a computationally motivated plug-in score, not the exact likelihood ratio of the complete SP model. A covariance obtained from the same observation is not independent secondary training data; classic adaptive-detection calibration cannot simply be imported.

For a preselected candidate in the projected Gaussian null, T_j is exponential with mean one. Selecting the maximum of M independent such scores gives

$$\Pr\{\max_j T_j > \tau\} = 1 - (1 - e^{-\tau})^M. \tag{25}$$

At a nominal single-test false-alarm level of 5% and $M = 8$, the selected maximum has false-alarm probability 33.66%. Real candidate scores are usually dependent, but dependence does not restore a generic single-test calibration. This matters when a scorer is interpreted as a posterior probability or a thresholded significance test rather than merely a ranking heuristic.

Verification. T07 checks projected likelihood, refitted residual decrease, and the rank-one Bayes formula against independent dense calculations. T15 obtains a selected false-alarm fraction 0.33482 from 100,000 independent null experiments, consistent with (25).

7 Data-aided channel estimation with a frozen working distribution

7.1 Exact second-moment objective

Let $q(d)$ be a fixed working distribution, possibly an exact codeword posterior or an approximation. Set

$$\mu = \mathbb{E}_q d, \quad V = \mathbb{E}_q[(d-\mu)(d-\mu)^H], \quad P_q = \mathbb{E}_q[(d-\mu)(d-\mu)^T]. \tag{26}$$

For active operators $U_1, \dots, U_S$, write

$$H(g) = \sum_{i=1}^{S} g_i U_i, \qquad m = p + \mu, \qquad \Phi_m = [U_1 m \ \cdots \ U_S m].$$

Proposition 8 (moment-consistent complete-data loss).

$$\mathbb{E}_q \|y - H(g)(p+d)\|^2 = \|y - \Phi_m g\|^2 + g^H \Gamma g, \qquad \Gamma_{ij} = \operatorname{tr}(U_i^H U_j V). \tag{27}$$

Furthermore $\Gamma \succeq 0$. With a proximal term $\lambda\|g - g_o\|^2$, $\lambda \geq 0$, a unique minimizer exists whenever the matrix below is positive definite and satisfies

$$(\Phi_m^H \Phi_m + \Gamma + \lambda I)\hat{g} = \Phi_m^H y + \lambda g_o. \tag{28}$$

Proof. Write $d = \mu + e$. The cross term in the squared norm vanishes because $\mathbb{E}_q e = 0$. The remaining uncertainty contribution is $\mathrm{tr}[H(g)^H H(g) V] = g^H \Gamma g$. If $V = RR^H$, then $g^H \Gamma g = \|H(g)R\|_F^2 \geq 0$. Differentiating the real quadratic objective with respect to g^* gives (28).

This identity does not require Gaussian q, independent symbols, or a circular posterior. The pseudo-covariance P_q does not independently enter this particular proper complex squared norm. It has a separate role in noncircular detection metrics, discussed below.

The uncertainty penalty is centered at zero. Increasing Γ changes the left-hand side of (28), but not its right-hand side. Increasing the proximal weight changes both sides. Treating these as the same regularization operation changes the objective.

If one chooses a fixed weighted squared norm with weight $W \succ 0$, its uncertainty term is $\Gamma_{ij}^W = \mathrm{tr}(U_i^H W U_j V)$. This is a valid different quadratic objective. It need not have the simplifications below and should not be silently described as the original white-thermal-noise complete-data likelihood.

Verification. T05 enumerates all 256 four-symbol QPSK vectors under a correlated working distribution. Equation (27) agrees to 1.42×10^{-14}. The normal-equation solution agrees with a separately constructed augmented least-squares system to 3.1×10^{-16} relative error.

7.2 The precise scalar-regularization special case

Proposition 9. Suppose $[U_i x]_n = u_i[n] x_{n-s_i}$, $|u_i[n]| = 1$, and the effective shifts s_i are distinct modulo N. If $V = \mathrm{diag}(v_0, \ldots, v_{N-1})$, then

$$\Gamma = \left(\sum_{n=0}^{N-1} v_n \right) I_S. \tag{29}$$

For general V,

$$\Gamma_{ij} = \sum_n u_i[n]^* u_j[n] V_{n-s_j, n-s_i}. \tag{30}$$

Proof. Expand $\mathbb{E}[(U_i e)^H (U_j e)]$ entrywise to obtain (30). With diagonal V, an off-diagonal matrix entry in (30) is zero unless $s_i = s_j$. Distinct shifts eliminate all cross-operator terms, and the diagonal terms equal $\sum_n v_n$.

Thus, under these conditions, a network that predicts N marginal variances and is consumed only by (28) affects the estimator through one scalar. Predicting a high-dimensional output does not create a high-dimensional channel-estimation correction at that interface. A fixed zero-centered ridge is an essential comparison.

The qualifications are material: two distinct integer delay-Doppler coordinates can have the same effective shift, and correlated V can produce nonzero off-diagonal Γ even with distinct shifts. Also, a physical fractional-path operator is generally not monomial in the DAFT domain. It must not

inherit (29) without rederivation. The project explicitly checked the noncollision condition for its restricted integer dictionary [P1].

Equation (30) identifies the required phase-weighted correlations. It does not prove they can be estimated at low cost. Posterior sampling, code constraints, lag storage, and effective sample size all remain part of the computation budget.

Verification. T06 checks (29), constructs a shift-collision counterexample with nonzero off-diagonal norm 1.00469, constructs a correlated-V counterexample, and verifies the lag expression (30).

7.3 How covariance error perturbs the gain update

Proposition 10 (fixed-right-hand-side sensitivity). Let

$$M = \Phi_m^H \Phi_m + \Gamma + \lambda I \succ 0, \quad b = \Phi_m^H y + \lambda g_o, \quad g = M^{-1} b.$$

If $\tilde{\Gamma} = \Gamma + E$ and $\|E\|_2 < \lambda_{\min}(M)$, then for $g \neq 0$,

$$\frac{\|\tilde{g} - g\|}{\|g\|} \leq \frac{\|E\|_2}{\lambda_{\min}(M) - \|E\|_2}. \tag{31}$$

Proof. Subtract the two systems: $\tilde{g} - g = -(M + E)^{-1} E g$. Bound the inverse norm using the smallest eigenvalue and $\lambda_{\min}(M + E) \geq \lambda_{\min}(M) - \|E\|_2$.

The bound conditions on the same Φ_m, b, g_o. Changes in posterior means or support require additional perturbation terms. The proposition is not a bound on true-channel estimation error or FER. T11 gives relative change 0.10486 below its bound 0.25.

7.4 A self-generated constraint can be algebraically inactive

Let $\hat{g}_{\mathrm{LS}}$ minimize $\|y - \Phi g\|^2$ and suppose a projector P_W is constructed so that $P_W \hat{g}_{\mathrm{LS}} = \hat{g}_{\mathrm{LS}}$. Then

$$\|y - \Phi g\|^2 + \lambda \|(I - P_W) g\|^2$$

has $\hat{g}_{\mathrm{LS}}$ as a minimizer for every $\lambda \geq 0$: the first term is already minimal there and the second equals zero. A hard restriction to range(P_W) has the same attainable minimum because it contains the LS solution. Uniqueness additionally requires suitable rank conditions.

This explains one project-specific null construction [P3]. It does not establish general equivalence between soft penalties and hard physical parameterization. T13 independently checks both constructions to numerical precision.

8 What exact Bayesian iteration guarantees

8.1 Exact joint formulation and EM

Let $P_{\mathcal{C}}(d)$ be the prior over valid coded data vectors. For a fixed support,

$$p(y, d \mid g) = P_{\mathcal{C}}(d) (\pi \sigma^2)^{-N} \exp[-\|y - H(g)(p + d)\|^2 / \sigma^2]. \tag{32}$$

Define the variational negative free energy, ignoring constants independent of q, g,

$$\mathcal{F}(q,g) = \frac{1}{\sigma^2}\mathbb{E}_q\|y - H(g)(p+d)\|^2 + D_{\mathrm{KL}}(q\|P_{\mathcal{C}}). \tag{33}$$

Its exact decomposition is

$$\mathcal{F}(q,g) = -\log p(y \mid g) + D_{\mathrm{KL}}(q\|p(d \mid y,g)) + \text{constant}. \tag{34}$$

Proposition 11 (the conditional EM guarantee). If the E-step uses the exact posterior $q_t = p(d \mid y, g_t)$, and the M-step does not increase the complete-data expected squared residual for that fixed q_t, then $p(y \mid g_{t+1}) \geq p(y \mid g_t)$.

Proof. At g_t, the KL term in (34) is zero. An M-step decreases $\mathcal{F}(q_t, g)$, and minimizing it over q at the new g cannot increase it. Equation (34) therefore implies nondecrease of the observed likelihood. This is the standard EM mechanism [R4]; (27) supplies the exact quadratic M-step for the linear coefficient representation.

An exact-posterior proximal M-step centered at g_t also gives a generalized EM descent if its total objective is no larger than the objective at g_t: its nonnegative proximal penalty implies the unpenalized expected residual has not increased. This statement does not apply automatically to arbitrary damping, changing supports, thresholding, or an approximate turbo distribution.

Three boundaries are essential. Exact EM can converge to a local solution. Likelihood improvement is not samplewise improvement in true channel NMSE. Neither likelihood improvement nor local convergence implies a monotone decrease of message-error risk for a finite-iteration receiver.

If q_t is only a detector/decoder approximation, (27) remains an exact identity for that chosen q_t, but the KL term in (34) is no longer zero at the current channel. Decreasing the frozen quadratic then need not improve the actual marginal likelihood. Moreover, even with exact q_t, dropping Γ changes the M-step.

Numerical counterexample. T10 uses a length-four, 16-codeword QPSK repetition model and exact enumeration of the code posterior. At a fixed observed block the initial log marginal likelihood is -22.64823. A mean-only channel update moves it to -30.56020, a decrease of 7.91197. The moment-consistent update including Γ moves it to -13.43058. The witness inputs are saved. A separate 20-step exact EM run is nondecreasing at every step.

8.2 Soft information, noncircularity, and normalization

For $z = x + n$, $n \sim \mathcal{CN}(0, \sigma_n^2)$, and uniform QPSK $x \in \{\pm a \pm ja\}$, take positive LLR to indicate the positive component. Then

$$L_I = 4a\,\mathrm{Re}\,z/\sigma_n^2, \qquad L_Q = 4a\,\mathrm{Im}\,z/\sigma_n^2. \tag{35}$$

Under independent I/Q bit probabilities, set $t_I = \tanh(L_I/2)$ and $t_Q = \tanh(L_Q/2)$. The exact moments of that working symbol distribution are

$$\mu = a(t_I + jt_Q), \quad v = a^2(2 - t_I^2 - t_Q^2), \quad c = a^2(t_Q^2 - t_I^2). \tag{36}$$

Even if the transmitted prior is circular, unequal bit reliabilities generally make the conditional symbol distribution noncircular. For general complex error e with variance $v = \mathbb{E}|e|^2$ and pseudo-variance $c = \mathbb{E}e^2$, its real covariance is

$$\Sigma_{\mathbb{R}} = \frac{1}{2}\begin{bmatrix} v + \operatorname{Re} c & \operatorname{Im} c \\ \operatorname{Im} c & v - \operatorname{Re} c \end{bmatrix}. \tag{37}$$

A nondegenerate complex Gaussian approximation requires $v > |c|$. Up to a normalization constant, its log density is

$$\ell(e) = -\frac{v|e|^2 - \operatorname{Re}(c^* e^2)}{v^2 - |c|^2}. \tag{38}$$

For independent interfering symbol errors, the observation-factor moments are

$$v_\eta = \sigma^2 + \sum_k |h_k|^2 v_k, \qquad c_\eta = \sum_k h_k^2 c_k,$$

excluding the target symbol's own contribution. Dividing by a complex channel gain h divides v by $|h|^2$ and c by h^2. These are classical noncircular inference constructions [R8], not new theory. Their usefulness in a particular finite graph still needs receiver-level verification.

Bit APPs must be obtained by marginalizing properly normalized finite-constellation weights including the applicable prior. Subtracting the target bit's own input LLR produces its extrinsic quantity. If I/Q are coupled, the other bit's prior still participates in marginalization and must not also be removed without derivation.

8.3 Moment-consistent damping

One explicit way to damp probability distributions is $q = (1-\eta)q_0 + \eta q_1$, $0 \le \eta \le 1$. Its moments satisfy

$$\begin{aligned} \mu &= (1-\eta)\mu_0 + \eta\mu_1, \\ V &= (1-\eta)V_0 + \eta V_1 + \eta(1-\eta)(\mu_1 - \mu_0)(\mu_1 - \mu_0)^H, \\ P_q &= (1-\eta)P_0 + \eta P_1 + \eta(1-\eta)(\mu_1 - \mu_0)(\mu_1 - \mu_0)^T. \end{aligned} \tag{39}$$

These follow from raw second moments. Damping only the mean while retaining variance around a different, undamped mean is not the moment set of this mixture. Equation (39) is one consistent damping policy, not a claim that every inference algorithm must use mixture damping. Natural-parameter damping defines a different policy and needs its own moment conversion.

T12 checks LLRs, moments, real/complex metrics, and mixture damping by exact finite-alphabet calculations. The example's omitted between-mean variance would be 0.12399 under naive variance averaging.

8.4 Extrinsic replacement is not statistical independence

For each decoder invocation, record its actual input likelihood and its APP output. Check-node/extrinsic output is formed by removing that invocation's input contribution. A new decoder invocation replaces the previous feedback prior rather than repeatedly adding the full APP to it.

This removes a specific explicit self-information duplication. It does not make messages on a loopy joint estimation/detection/decoding graph independent, and it does not turn an approximate symbol distribution into the true codeword posterior. The September project reassessment makes this same distinction [P1, Main Derivation 6]. An additional decoder call also consumes additional computation [R6].

8.5 Why residual energy can be below the thermal noise variance

Proposition 12 (known-channel posterior variance identity). For known H and the exact data posterior, let $\mu(y) = \mathbb{E}[d \mid y]$ and $V(y) = \mathrm{Cov}(d \mid y)$. Then

$$N\sigma^2 = \mathbb{E}\|y - H(p + \mu(y))\|^2 + \mathbb{E}\,\mathrm{tr}[HV(y)H^H]. \tag{40}$$

Proof. Since $w = y - H(p+d)$, its conditional mean is $y - H(p+\mu(y))$ and its conditional covariance is $HV(y)H^H$. Apply the law of total variance and take the trace.

Thus the same-observation posterior-mean residual is generally smaller in expected energy than the original noise. This is not by itself a real-versus-complex factor-of-two error. Equation (40) is an expectation over observations, not a per-frame identity. It does not grant unbiasedness to unknown-channel fitting or approximate posterior moments. In particular, after conditioning on y, d and w are dependent; one must not automatically add $HV(y)H^H$ and $\sigma^2 I$ as if they were independent posterior disturbances.

T18 uses deterministic Gauss-Hermite quadrature for a known scalar channel and exact QPSK posterior. Residual energy 0.3006896363 plus posterior uncertainty energy 0.1993103637 equals thermal variance 0.5 to 1.1×10^{-16}.

9 Connecting channel representation to communication risk

9.1 The channel error metric induced by SP transmission

Let H be a fixed actual channel and G a fixed approximation. Write $\Delta H = H - G$, $\mu_0 = \mathbb{E}d$, $V_0 = \mathrm{Cov}(d)$, and

$$R_x = \mathbb{E}[(p+d)(p+d)^H] = V_0 + (p+\mu_0)(p+\mu_0)^H.$$

The mean observation error caused by using G is

$$\begin{aligned} \mathcal{J}_H(G) &= \mathbb{E}\|\Delta H(p+d)\|^2 \\ &= \|\Delta H(p+\mu_0)\|^2 + \mathrm{tr}(\Delta H V_0 \Delta H^H) \\ &= \mathrm{tr}(\Delta H^H \Delta H R_x). \end{aligned} \tag{41}$$

For zero-mean isotropic data this is $\|\Delta Hp\|^2 + P_d\|\Delta H\|_F^2$. The pilot-sensitive term cannot generally be replaced by a multiple of Frobenius error for a fixed pilot. If one additionally averages over independent zero-mean isotropic random pilots, a scalar multiple results, but that is a different ensemble.

Equation (41) measures observation mismatch, not FER directly. Nevertheless, it supplies a rigorous bridge to risk under explicit fixed-model assumptions.

9.2 A model-error divergence identity and risk-transfer bounds

Let P_H and P_G denote the joint laws of (b, y) with the same message prior, codebook, pilot, and proper noise variance, differing only in the fixed channel operator. Natural logarithms are used.

Proposition 13 (observation mismatch equals joint-model divergence).

$$D_{\mathrm{KL}}(P_H\|P_G) = \frac{\mathcal{J}_H(G)}{\sigma^2}. \tag{42}$$

For any one fixed receiver decision map $\mathcal{A}$, used under both propagation laws,

$$|R_H(\mathcal{A}) - R_G(\mathcal{A})| \le \min\left\{1, \sqrt{\frac{\mathcal{J}_H(G)}{2\sigma^2}}\right\}. \tag{43}$$

Let δ_G^* be a message-MAP rule optimal under P_G and δ_H^* a message-MAP rule optimal under P_H. Then

$$0 \le R_H(\delta_G^*) - R_H(\delta_H^*) \le \min\left\{1, \sqrt{\frac{2\mathcal{J}_H(G)}{\sigma^2}}\right\}. \tag{44}$$

Proof. Conditional on a message, the two observations are proper Gaussian with identical covariance $\sigma^2 I$. Their KL divergence equals the squared distance between their means divided by σ^2. Average over the common message prior to obtain (42).

Define total variation by $\mathrm{TV}(P, Q) = \sup_E |P(E) - Q(E)|$, equal to half the L^1 distance of their densities. The error event of a fixed receiver is an event in the joint (b, y) space, so its probability difference is at most TV. Pinsker's inequality gives $\mathrm{TV}(P, Q) \le \sqrt{D_{\mathrm{KL}}(P\|Q)/2}$ and proves (43). For completeness, partitioning the sample space into the positive-density-difference set and its complement reduces KL to Bernoulli KL by the log-sum inequality. For fixed Bernoulli parameter q, the second derivative with respect to p is $1/[p(1-p)] \ge 4$. Taylor's theorem about its minimum at $p = q$ yields $D_{\mathrm{KL}} \ge 2(p-q)^2 = 2\,\mathrm{TV}^2$.

Finally,

$$\begin{aligned} R_H(\delta_G^*) - R_H(\delta_H^*) = {} & [R_H(\delta_G^*) - R_G(\delta_G^*)] \\ & + [R_G(\delta_G^*) - R_G(\delta_H^*)] \\ & + [R_G(\delta_H^*) - R_H(\delta_H^*)]. \end{aligned}$$

The middle term is nonpositive by optimality under G, and each remaining term is bounded by TV. This proves (44). The inequalities are general probabilistic bounds, not an AFDM-specific optimality claim; [R9] provides background on the Pinsker relation.

Meaning and limitations. Equation (43) rigorously bounds the error introduced by substituting one propagation model for another while keeping the receiver rule fixed. Equation (44) bounds excess risk for an exact matched-model Bayes receiver. Neither inequality establishes a monotone ordering of actual FER when $\mathcal{J}$ changes, and neither certifies approximate BP as Bayes-optimal. The bounds can be vacuous at practical mismatch levels; this happens for the aggressively truncated toy models below.

The fixed-model qualification is crucial. One must not insert a channel estimate $G(y)$ learned from the same observation into a Gaussian generative model with fixed mean and then apply (42) as though G were exogenous. With independent training, conditioning on that training can justify an analogous conditional argument. For a prespecified alternative propagation model, a fixed receiver map may internally perform adaptive estimation; (43) still compares that same complete map under the two laws. These are distinct uses of the result.

9.3 Weighted channel projection is not Frobenius projection

For a fixed subspace $\mathrm{span}(U_i : i \in \mathcal{S})$, minimizing (41) over $G(g)$ gives

$$M^R_{ij} = \mathrm{tr}(U_i^H U_j R_x), \qquad b^R_i = \mathrm{tr}(U_i^H H R_x), \qquad M^R g = b^R, \tag{45}$$

with a pseudoinverse if the weighted problem is singular. This is a diagnostic projection involving the true H, not a deployable estimator.

Take $H = \mathrm{diag}(1, 3)$, restrict $G = gI$, choose $p = (2, 0)^T$ and $V_0 = 0.5I$. The Frobenius minimizer is $g_F = 2$, with Frobenius error 2 and observation-mismatch energy $\mathcal{J} = 5$. The SP-weighted minimizer is $g_R = 1.2$, with larger Frobenius error 3.28 but smaller observation mismatch $\mathcal{J} = 1.8$. T09 verifies these numbers exactly.

The usual representation/coefficient decomposition remains valid for a Frobenius-orthogonal projection:

$$\|H - \widehat{H}\|_F^2 = \|H - H^F_{\mathcal{S}}\|_F^2 + \|H^F_{\mathcal{S}} - \widehat{H}\|_F^2, \quad \widehat{H} \in \mathrm{span}(U_i : i \in \mathcal{S}). \tag{46}$$

It is not a decomposition of FER. This distinction agrees with the September project result in which a true-data LS intervention produced fewer failed frames than the Frobenius-best projection on the same support [P1, Main Derivation 4]. That observation does not violate (46); it evaluates a different loss through a particular receiver.

For a single path and random isotropic pilot/data averaging, (42) reduces to $(P_p + P_d)N|h|^2\epsilon_{\mathcal{K}}/\sigma^2$. Thus a small relative operator-energy tail need not produce a small whole-block divergence. With a fixed pilot, its particular alignment contributes separately through (41).

10 A theoretically consistent, integer-domain receiver construction

10.1 Architecture and status

The following construction specifies a method whose local operations correspond to stated objectives. It is a **theory-consistent reference design**, not a claim of a newly superior I1 receiver. It retains integer channel coefficients and does not introduce continuous Doppler estimation.

Use full fractional-Doppler propagation for the physical reference experiment. Restrict only the receiver representation to a declared integer dictionary and active budget. Initialize from the scaled pilot, using an existing qualified method such as the project's OMP-based initializer. Maintain the established I1 implementation as a separate frozen benchmark rather than silently replacing it with this construction.

At each permitted outer iteration, obtain detector and, when scheduled, decoder messages through documented extrinsic interfaces. Form a declared working distribution or approximate moment set. Refit the active channel using (28). Use (29) when its conditions hold; otherwise specify the richer Γ and include the acquisition cost of its required correlations. Choose one explicitly defined damping policy and reconstruct the operator consistently.

An optional support step uses a finite candidate pool and compares complete, refitted, frozen-q objectives, including the no-change alternative. It is never required to insert or protect a new atom when no improving action exists. If a support coordinate is removed, it must not persist silently in a separately damped old channel matrix.

Decoder refresh schedules, maximum iteration counts, candidate budgets, and thresholds are frozen before an independent comparison. The exact formulas alone do not select those design constants. Because project tests already found no net advantage for one related support-exchange implementation, a new implementation must be judged against that negative evidence, not presented as a success by construction [P1].

10.2 Exact fixed-objective add/drop calculations

Define a global integer coordinate space for the current candidate pool. With fixed q, λ, g_o, use

$$J(g) = \|y - \Phi_m g\|^2 + g^H \Gamma g + \lambda \|g - g_o\|^2 = c_0 - 2\,\mathrm{Re}(g^H b) + g^H M g, \tag{47}$$

where $M = \Phi_m^H \Phi_m + \Gamma + \lambda I$, $b = \Phi_m^H y + \lambda g_o$, and $c_0 = \|y\|^2 + \lambda \|g_o\|^2$. Assume the principal blocks being used are positive definite. On support $\mathcal{S}$,

$$J_{\mathcal{S}}^* = c_0 - b_{\mathcal{S}}^H M_{\mathcal{S}\mathcal{S}}^{-1} b_{\mathcal{S}}. \tag{48}$$

Proposition 14 (Schur-complement support comparison). For a proposed addition $j \notin \mathcal{S}$, write $a = M_{\mathcal{S}j}$,

$$s_j = M_{jj} - a^H M_{\mathcal{S}\mathcal{S}}^{-1} a > 0, \qquad t_j = b_j - a^H \hat{g}_{\mathcal{S}}.$$

Then

$$J^*_{\mathcal{S}} - J^*_{\mathcal{S}\cup j} = \frac{|t_j|^2}{s_j}. \tag{49}$$

For an augmented support $\mathcal{E}$ with optimizer $\hat{g}_{\mathcal{E}}$, forcing coordinate i to zero and refitting the others increases the optimum by

$$J^*_{\mathcal{E}\setminus i} - J^*_{\mathcal{E}} = \frac{|\hat{g}_{\mathcal{E},i}|^2}{(M^{-1}_{\mathcal{E}\mathcal{E}})_{ii}}. \tag{50}$$

Proof. Eliminate the existing coefficients in the block quadratic. The remaining scalar quadratic has curvature s_j and linear term t_j, giving (49). For (50), impose the linear constraint $e_i^H g = 0$ on the strictly convex quadratic. Its Lagrange-multiplier solution has increase $|e_i^H \hat{g}|^2/(e_i^H M^{-1} e_i)$.

These formulas can evaluate a family of add/drop swaps without solving every enlarged system from scratch. Practical implementations should use Cholesky updates and triangular solves; explicit inverses appear here only to express the identities. T14 compares all 20 swaps in a small example against complete independent refits. The largest discrepancy is 1.78×10^{-15}; the best objective changes from 12.03334 to 11.39852.

A drop must retain the regularization constant of a removed old coordinate. Expressing all supports in the same global objective, as in (47), makes this automatic. Omitting the term $\lambda|g_{o,i}|^2$ when dropping i changes the comparison and can reward removal for the wrong reason.

Accepting a swap only when (48) decreases guarantees **frozen-objective descent**. It does not guarantee FER descent or marginal-likelihood ascent with approximate q. Repeated support search over noisy scores also remains subject to selection effects.

10.3 Computation accounting

For integer monomial atoms, applying a channel with S active coefficients costs $O(NS)$. Applying a physical P-path channel between DAFT vectors can use $O(PN + N \log N)$ work through time-domain paths and transforms. This does not imply equally cheap path estimation or decoding. Also, independent DAFT-domain QAM does not remain an independent QAM prior after transformation to time domain.

Operation	Straightforward cost	Important qualification
Form active data-aided columns	$O(NS)$	Their input moments must already be available.
Form active Gram matrix	$O(NS^2)$	Structure or incremental updates may reduce this.
Dense active coefficient solve	$O(S^3)$	Positive definiteness and conditioning must be checked.
Diagonal-V, noncollision Γ	$O(N + S)$	Only the total variance is consumed.
General Γ from accessible V	$O(NS^2)$ using (30)	Obtaining or storing V is additional, potentially dominant work.
Candidate cross products	$O(NS)$ per candidate	Plus $O(S^2)$ triangular solves/updates.

Operation	Straightforward cost	Important qualification
Dense covariance whitening	$O(N^3)$ factorization, $O(N^2)$ storage	A cheap covariance matvec is not a cheap exact inverse.
Sparse detector sweep	Proportional to graph edges and alphabet work	More retained leakage coefficients increase graph size.
Decoder refresh	Implementation- and code-dependent	Every extra invocation must be counted.

Exact code-posterior enumeration grows with codebook size and is only a small-system reference. A small $S \times S$ output statistic does not make posterior acquisition inexpensive. Low-rank, lag-restricted, sampled, and learned covariance approximations must each be charged for their inputs and validation.

The complete cost remains

$$C_{\text{total}} = C_{\text{init}} + \sum_t (C_{\text{CE},t} + C_{\text{det},t} + C_{\text{moments},t}) + \sum_r C_{\text{dec},r} + C_{\text{support}}. \tag{51}$$

Measured percentages from an old implementation do not transfer to this new reference script or another device. The script's execution time is not a production receiver timing result.

11 Numerical verification and controlled exploration

11.1 Reproducibility contract

All new results are produced by `verify_theory.py` using NumPy complex128 and SciPy. The master seed is 20260919; experiment-specific streams are fixed in the code. The script writes numerical assertions, result tables, and raw witness arrays. A separate saved-array audit independently recomputes every saved toy decision and the EM witness without importing the simulator; all 23 audit assertions pass. A second simulator run in the same environment reproduces all 31 raw arrays exactly and all JSON values except elapsed time. No project test set, production code, external server, or hidden channel estimate is used.

The 50 assertions are organized into T01-T20. Repeated parameter settings and multiple assertions within a group are not independent scientific replications. Deterministic identities are checked against independently constructed expressions; stochastic equalities use their Monte Carlo standard errors. The largest deterministic/finite-enumeration discrepancy is 1.42×10^{-14}. Such agreement checks implementation consistency, while the proofs establish the stated universal finite-dimensional results.

Groups	Verified object	Method
T01-T02, T20	Prefix, DAFT, integer expansion, orthogonality, merged multipath tails	Direct matrices, analytic entries, FFT action, trace identities
T03-T04	Conditional covariance, white single path, non-Gaussianity	Exact QPSK enumeration and closed fourth-moment formula
T05-T06	Frozen-q loss, Γ, collapse, collision and correlation exceptions	Correlated finite-alphabet enumeration and augmented least squares
T07, T15	Refitted evidence, OMP/OLS distinction, selection effects	Exact refits/Bayes factors; 100,000 null experiments

Groups	Verified object	Method
T08-T09	Tail cross covariance and pilot-aware channel error	Independent operator and trace computations
T10-T14	Exact EM, sensitivity, moments, inactive constraints, support updates	Exact codeword sums, counterexamples, complete refits
T16	Propagation-model mismatch, exact decoding, KL identity	40,000 paired finite-code experiments
T17-T18	Flat-SP MSE and posterior noise identity	100,000 blocks; deterministic quadrature
T19	Sign of a reported project factorial contrast	Arithmetic on explicitly attributed counts, not raw replay

11.2 Exact-decoding toy experiment: full propagation versus truncated propagation

To isolate representation from approximate inference, use $N = 8$, $c_1 = 3/(2N)$, $c_2 = \sqrt{2}/N^2$, one path with integer delay $l = 1$, Doppler $\nu = 0.5$, and unit gain. Two independent QPSK symbols from $\{\pm 0.5 \pm 0.5j\}$ are repeated four times, producing a 16-codeword length-eight codebook. It carries four information bits in sixteen QPSK-label positions, so its binary code rate is 1/4. It is **not LDPC**.

The pilot is $p_n = \sqrt{0.5}e^{j0.41n}$, and complex thermal noise variance is 0.3. Each trial draws a fresh equiprobable message and independent noise. The physical channel is fixed; the experiment does not average over channel geometries. The code induces correlated data, and its exact second moment is used in (42).

Construct G_K from the K largest integer coefficients of the exact full path, without renormalization. All coefficients are known exactly. The receiver exhaustively compares all sixteen codeword likelihoods using G_K, so it is exact ML/MAP for that model, not an approximate detector or channel estimator.

For each K, compare two experiments using the same messages and noise: actual full propagation $y = H(p + d) + w$ decoded with G_K, and artificially truncated propagation $y_K = G_K(p + d) + w$ decoded with that same G_K. This comparison asks whether testing a receiver only under a matching truncated generator can misrepresent its performance under the full channel.

K	Full propagation: failures / 40,000	Full-propagation FER	Truncated-propagation FER	Full minus truncated FER
1	22,166	0.554150	0.183750	+0.370400
2	4,155	0.103875	0.070175	+0.033700
3	4,010	0.100250	0.054200	+0.046050
5	1,236	0.030900	0.041350	-0.010450
8	753	0.018825	0.018825	0

For example, the marginal 95% Clopper-Pearson interval for the full-propagation $K = 2$ result is $[0.100901, 0.106906]$. Complete intervals, paired differences, standard errors, and counts are saved in `coded_toy.csv`. These are descriptive intervals for this fixed experiment, not multiplicity-adjusted confirmation of a selected new receiver. The same 40,000 independent message/noise trials are shared across rows; the number of independent trials is not multiplied by the number of configurations.

At $K = 1, 2, 3$, the matching truncated generator is optimistic. At $K = 5$, it is pessimistic. Thus a matched truncated simulation is not a universal upper or lower performance bound for full propagation. Removing tails changes both mismatch and the signal geometry/energy; retaining identical path gain and noise does not eliminate that change.

The $K = 8$ row is the full-channel reference for the same code and noise. Finite-sample observations need not strictly preserve expected Bayes-risk order: a tiny perturbation $G = H + 0.015I$ produces 752 observed failures versus 753 for the full model, which is not evidence of superiority over the true-model MAP rule.

11.3 Numerical interpretation of the risk bound

For $G = H + 0.015I$ in the same toy model,

$$D_{\mathrm{KL}}(P_H \| P_G) = 0.006,$$

verified both from codeword means and the covariance-trace expression. Equation (43) gives a fixed-receiver risk-transfer bound $\sqrt{0.006/2} = 0.05477$. Equation (44) gives a mismatched-Bayes excess-risk bound $\sqrt{2(0.006)} = 0.10954$. The bounds are valid but conservative; their role is rigorous control, not accurate FER prediction.

For the larger truncation mismatches, the corresponding bounds saturate at one and are uninformative. Reporting only the existence of a bound without disclosing its numerical looseness would misrepresent the result. Direct Monte Carlo averages of log likelihood ratios agree with the analytic KL values within the specified standard-error checks.

11.4 What these experiments do and do not establish

The experiments establish implementation consistency of the operators and local objectives, explicit counterexamples to several overstrong claims, and a reproducible small-system demonstration that fractional-tail modeling affects coded decisions even with exact finite-code decoding.

They do not establish the best expansion width for the production system, performance gains from the proposed moment/support updates, reliable high-dimensional code-posterior acquisition, generalization over channel/LDPC families, or a production complexity advantage. In particular, they do not close the September report's outstanding full-G1 production comparison [P1-P2]. They close a smaller and well-specified reference-model question.

12 Reconciliation with project evidence and thesis claims

12.1 Corrections that belong in the theoretical section

Whiteness is not Gaussianity. A white single-path covariance is compatible with strongly non-Gaussian QPSK interference. Large N alone does not force a central-limit regime; Proposition 6 supplies an explicit counterexample.

Refitted likelihood is not ordinary OMP. The exact fixed-covariance candidate test after eliminating existing unknown coefficients is OLS-type, with projected normalization. A Bayesian posterior-odds score requires additional prior and determinant terms. Blanket claims that every version of Λ reduces to OMP should be replaced by the precise special cases in Section 6.

A frozen-q identity is not a complete receiver guarantee. The Γ identity is exact; a turbo approximation to q is not thereby exact; and lower frozen loss is not lower FER. This is consistent with the more cautious September reassessment [P1], rather than the strongest wording in earlier summaries.

A truth intervention is not always a performance bound for an arbitrary implementation. True channel/data inputs can be supplied through different schedules or interfaces. Their observed ordering does not establish a new information-theoretic limit. In particular, an oracle containing true hard symbols cannot underperform because those true symbols contain hard-decision errors; any explanation must instead examine the algorithms, scheduling, and other changed inputs.

Null results do not prove universal impossibility. No observed advantage of one BP iteration extension, whitening approximation, or pilot design does not exclude all alternatives. A confidence interval containing zero is not by itself an equivalence test. The production evidence should be reported at its actual algorithm, parameter, sample, and device scope.

12.2 Correcting the additive FER interaction sign

The September 11 summary [P3, Section 4] reports, at 8 dB, four conditions with failure counts $(37, 7, 3, 0)$ out of 256 frames: both channel uses estimated, true pilot cancellation only, true data channel only, and both true. Let the corresponding risks be $r_{00}, r_{10}, r_{01}, r_{11}$.

The raw risk interaction is

$$r_{11} - r_{10} - r_{01} + r_{00} = \frac{27}{256} = 0.10546875. \tag{52}$$

However, the extra joint *risk reduction* beyond adding the two individual reductions is

$$(r_{00} - r_{11}) - [(r_{00} - r_{10}) + (r_{00} - r_{01})] = -\frac{27}{256}. \tag{53}$$

Thus the benefits are subadditive/overlapping on the additive FER scale, not positively superadditive on that scale. The individual interventions are still beneficial. Different risk transforms define different interaction measures, but their scale must be stated. T19 checks this arithmetic only; it does not recompute the report's confidence interval or reconstruct the paired raw experiment.

12.3 What can now be finalized

Under the declared sampled-channel assumptions, the following pieces form a consistent theoretical section: the exact CPP/DAFT input-output model; fractional-Doppler expansion with coherent merging and explicit tails; conditional nuisance moments and non-Gaussianity limits; projected support evidence; frozen-distribution CE and its scalar special case; exact versus approximate EM; proper/nonproper soft information; pilot-sensitive model error and conditional risk-transfer bounds; and objective-consistent support update formulas with full computation accounting.

The contribution claim should remain narrower than "we solve SP-AFDM reception." The chapter establishes a mathematically consistent formulation and identifies exactly which approximation each practical block makes. It supplies a candidate construction and reproducible verification, but does not claim an empirically superior complete receiver or novelty for classical constituent formulas. Direct

SP-AFDM fractional-parameter SBL work already occupies much of the broad method combination [R7].

Three receiver-level questions remain open in the available evidence: whether a useful approximation to the relevant code-posterior statistics can be obtained at lower total cost; whether a representation/update policy beats frozen I1 and suitable classical controls on independent full-propagation trials; and whether that advantage persists across channel, pilot-power, and code configurations. These are concrete empirical obligations, not gaps hidden inside an asserted theorem.

13 References and source record

13.1 Primary public literature

[**R1**] A. Bemani, N. Ksairi, and M. Kountouris, "Affine Frequency Division Multiplexing for Next Generation Wireless Communications," *IEEE Transactions on Wireless Communications*, 22(11), 8214-8229, 2023. DOI: 10.1109/TWC.2023.3260906. Author manuscript: arXiv:2204.12798.

[**R2**] K. Zheng, M. Wen, T. Mao, L. Xiao, and Z. Wang, "Channel Estimation for AFDM With Superimposed Pilots," *IEEE Transactions on Vehicular Technology*, 74(2), 3389-3394, 2025. DOI: 10.1109/TVT.2024.3469380. Author manuscript: arXiv:2404.10232.

[**R3**] F. Huang, Q. Guo, Y. Zhang, and Y. Zakharov, "Message Passing-Based Joint Channel Estimation and Signal Detection for OTFS with Superimposed Pilots," *IEEE Transactions on Vehicular Technology*, 2024. DOI: 10.1109/TVT.2024.3378833. Author manuscript: arXiv:2309.08177v2.

[**R4**] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum Likelihood from Incomplete Data Via the EM Algorithm," *Journal of the Royal Statistical Society, Series B*, 39(1), 1-22, 1977. DOI: 10.1111/j.2517-6161.1977.tb01600.x.

[**R5**] C. Soussen, R. Gribonval, J. Idier, and C. Herzet, "Joint k-Step Analysis of Orthogonal Matching Pursuit and Orthogonal Least Squares," *IEEE Transactions on Information Theory*, 59(5), 3158-3174, 2013. DOI: 10.1109/TIT.2013.2238606. Author manuscript: arXiv:1111.0522.

[**R6**] Y. Kanazawa, H. Iimori, C. Pradhan, S. Malomsoky, and N. Ishikawa, "Superimposed Pilot-Based OTFS: Will It Work?," *IEEE Transactions on Vehicular Technology*, 2026. DOI: 10.1109/TVT.2025.3608199. Author manuscript: arXiv:2501.15935v3, revised 24 February 2026.

[**R7**] Y. Luo, Y. L. Guan, Y. Ge, Y. Jiang, L. Meng, and D. González G, "Data-aided Channel Estimation and Sensing With Sparse Bayesian Learning for AFDM-ISAC System," arXiv:2607.18881v1, 21 July 2026. Used as direct recent primary prior art; its numerical results are not reproduced here.

[**R8**] S. Yang, L. Wang, T. Lv, and L. Hanzo, "Approximate Bayesian Probabilistic-Data-Association-Aided Iterative Detection for MIMO Systems Using Arbitrary M-ary Modulation," *IEEE Transactions on Vehicular Technology*, 62(3), 1228-1240, 2013. DOI: 10.1109/TVT.2012.2227863. Author manuscript: arXiv:1311.1712.

[**R9**] D. Berend, P. Harremoës, and A. Kontorovich, "Minimum KL-divergence on complements of L1 balls," arXiv:1206.6544, 2012. Background on the divergence/total-variation relation; the bound used here is also proved in Section 9.

13.2 Project documents consulted

[**P1**] `l16-reassessment-v1.md`, "I1 lightweight diagnostics: derivation and research reassessment," September 12, 2026 batch. Retrieved in full through the Project/Library file interface. Main source for the current G0 assumptions, Gamma identities, negative support-update results, feedback limitations, and outstanding production validation.

[**P2**] `final-report-template-v1.md`, "I1 lightweight gap closure: results, derivation and research decisions," September 12, 2026 batch. A report template containing unfilled table placeholders, not a substitute for its raw result arrays. Retrieved in full; its listed remote arrays were not accessed.

[**P3**] `afdm_comprehensive_supplement_latest_findings(1).md`, September 11, 2026. Supplied historical summary. Used for explicitly attributed factorial counts and conditional regularization examples; overbroad interpretations are not adopted.

[**P4**] `AFDM_claim_reconstruction_audit_20260726(1).html`, July 26, 2026. Supplied historical audit. Used to identify earlier scoring, projection, Gaussian-approximation, and claim-boundary issues, not to certify new production performance.

Public primary sources were checked on September 19, 2026. Project documents are snapshots. This chapter's simulations and derivations were created for the present verification task; they are not measurements made by the authors of the cited project reports.